

CUTANEOUS DISORDERS OF DIABETES MELLITUS ASSOCIATED WITH METABOLIC, VASCULAR, NEUROLOGIC, OR IMMUNOLOGIC ABNORMALITIES

Acanthosis Nigricans (See Chap. 154)

EPIDEMIOLOGY

Acanthosis nigricans is one of the most recognizable cutaneous manifestations of diabetes. It is common in the general population, with the majority of cases associated with obesity and insulin resistance. In some instances, increased androgen production may also contribute.[16,17] Drug-induced, idiopathic, and familial forms of acanthosis nigricans have been reported.[18]

Acanthosis nigricans is considered a clinical marker for insulin resistance and a prognostic indicator for the development of type 2 diabetes.[19] In a large population-based study conducted in Galveston, Texas, acanthosis nigricans was present in 7% of school-aged children. This prevalence increased to 66% among children who weighed 200% of their ideal body weight. Fasting insulin levels in this cohort correlated directly with the presence and severity of skin changes.

The prevalence of acanthosis nigricans varies across ethnic groups. In the Galveston study, despite similar obesity rates, the condition was less common in white children (0.5%) and Hispanic children (5%) compared to African American children (13%).^[17] This suggests a possible genetic predisposition or increased cutaneous sensitivity to hyperinsulinemia in certain populations.^[19]

Although acanthosis nigricans has historically been linked to malignancy (particularly gastric adenocarcinoma), this association is rare. Evaluation for underlying malignancy is only warranted in cases with sudden onset, severe clinical presentation, or in non-obese, non-diabetic adults. Routine age-appropriate cancer screening is otherwise sufficient. In one clinician's experience evaluating over 12,000 cancer patients, only two cases of acanthosis nigricans were observed.^[20]

ETIOLOGY AND PATHOGENESIS

Insulin plays a central role in the development of acanthosis nigricans. Insulin resistance leads to hyperinsulinemia, which in turn stimulates insulin-like growth factor-1 (IGF-1) receptors in the skin. This cross-activation promotes keratinocyte and fibroblast proliferation, resulting in the characteristic epidermal thickening, hyperpigmentation, and velvety texture of affected skin.

Dysregulated lipid metabolism in insulin-deficient diabetes further contributes to cutaneous manifestations. Lipoprotein lipase (LPL) activity is insulin-dependent,^[13] and its deficiency in uncontrolled diabetes impairs clearance of triglyceride-rich chylomicrons and very-low-density lipoproteins (VLDL), leading to severe hypertriglyceridemia. This can manifest clinically as eruptive xanthomas.

Macro- and microangiopathy are key contributors to diabetic skin complications. Diabetic microvasculature exhibits increased permeability, reduced responsiveness to sympathetic stimulation, and impaired adaptation to thermal and hypoxic stress.^[10] These abnormalities, combined with large-vessel arteriosclerosis, predispose to poor wound healing and the development of diabetic foot ulcers.

Additionally, peripheral sensory neuropathy results in loss of protective sensation, increasing the risk of trauma and infection. The disruption of neuroinflammatory signaling further impairs tissue repair, contributing to chronic, non-healing ulcers.[14] Patients with diabetes who lack vibratory perception in the lower extremities have a 15.5-fold higher risk of leg amputation.[15]

Disorders of glucose and insulin signaling thus underlie a wide spectrum of cutaneous conditions through metabolic, vascular, neurologic, and immunologic mechanisms.